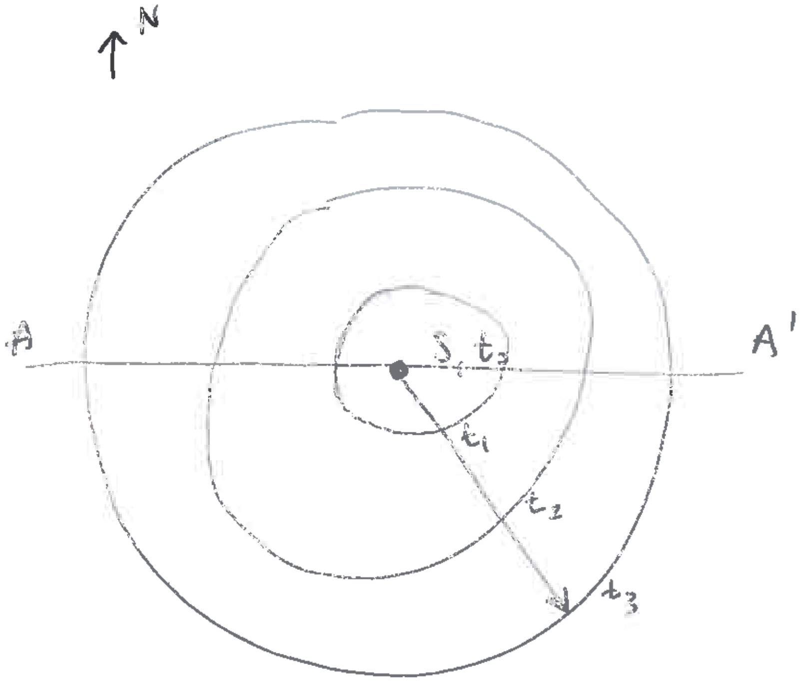
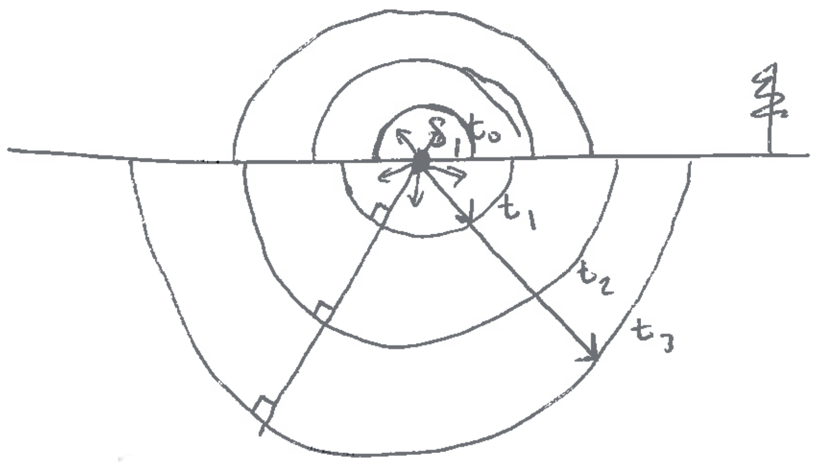
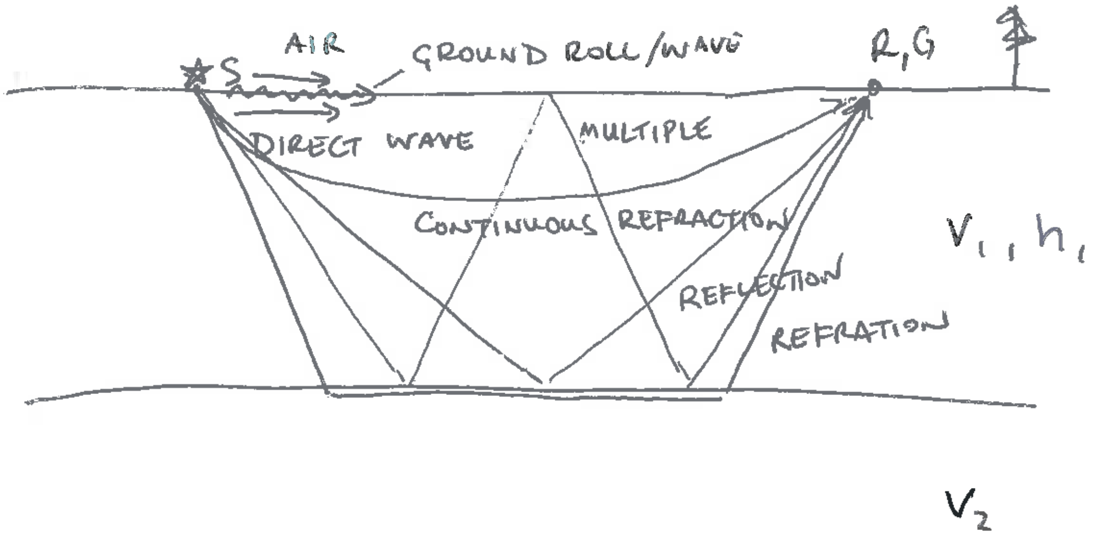
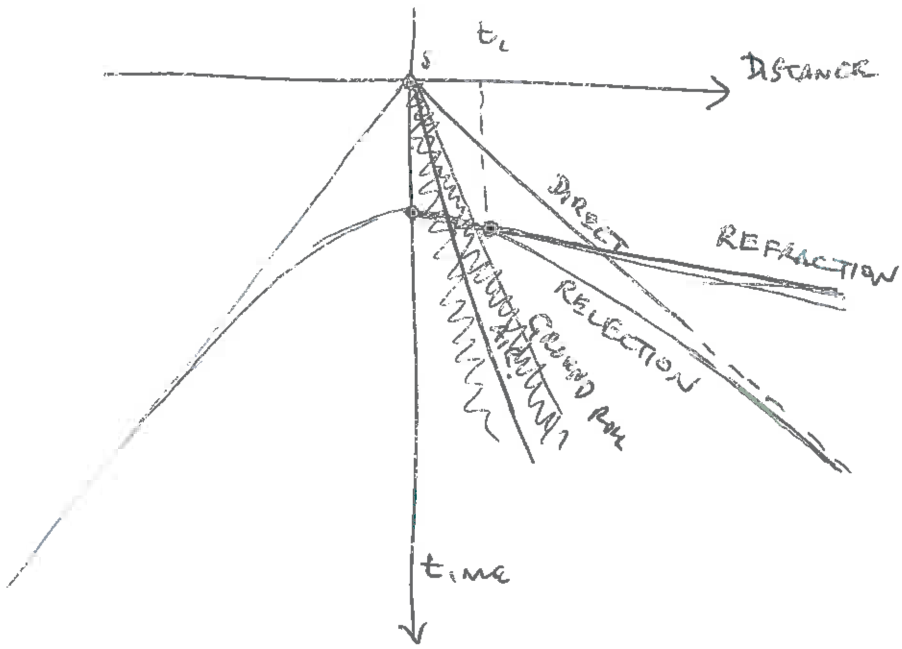
MAP VIEW





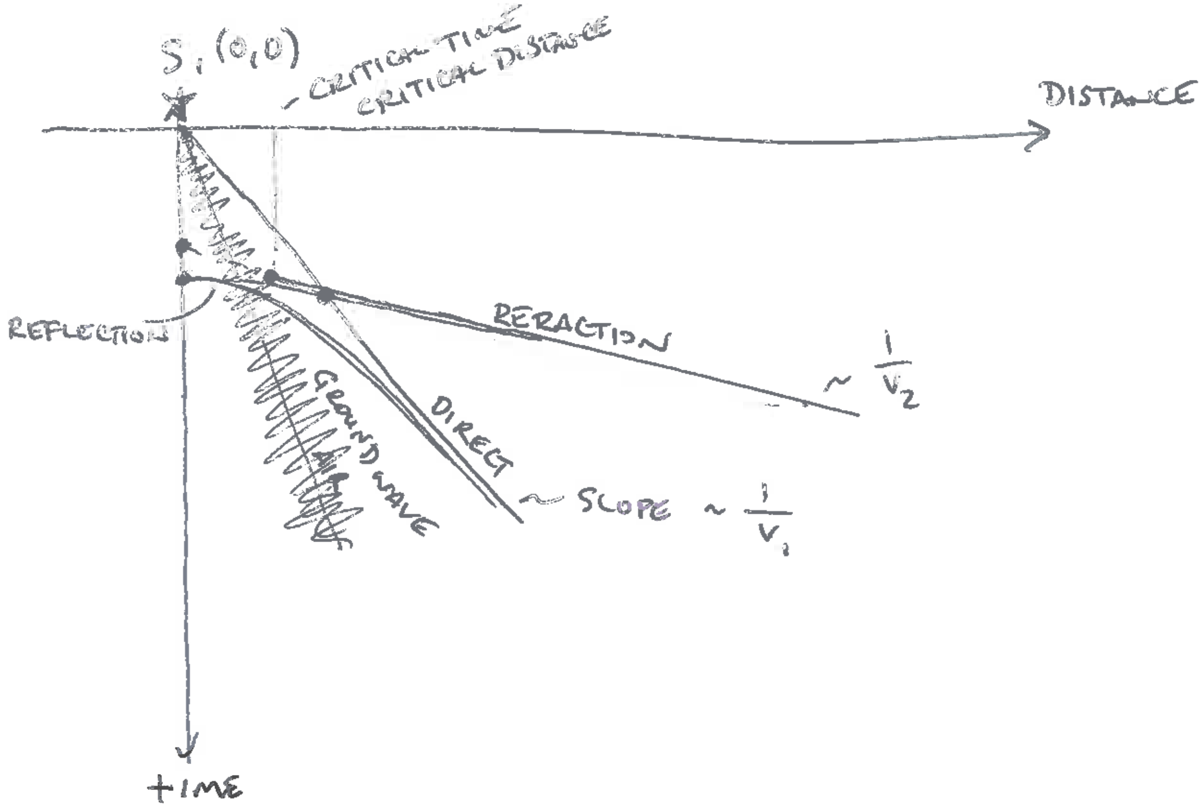
$$l_{WAVEFRONT} = t_{ELAPSED} V_{MEDIUM}$$

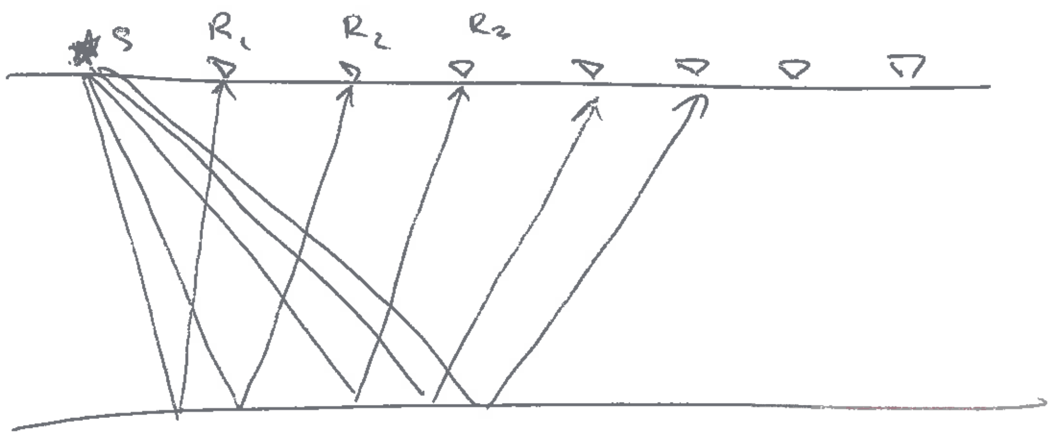


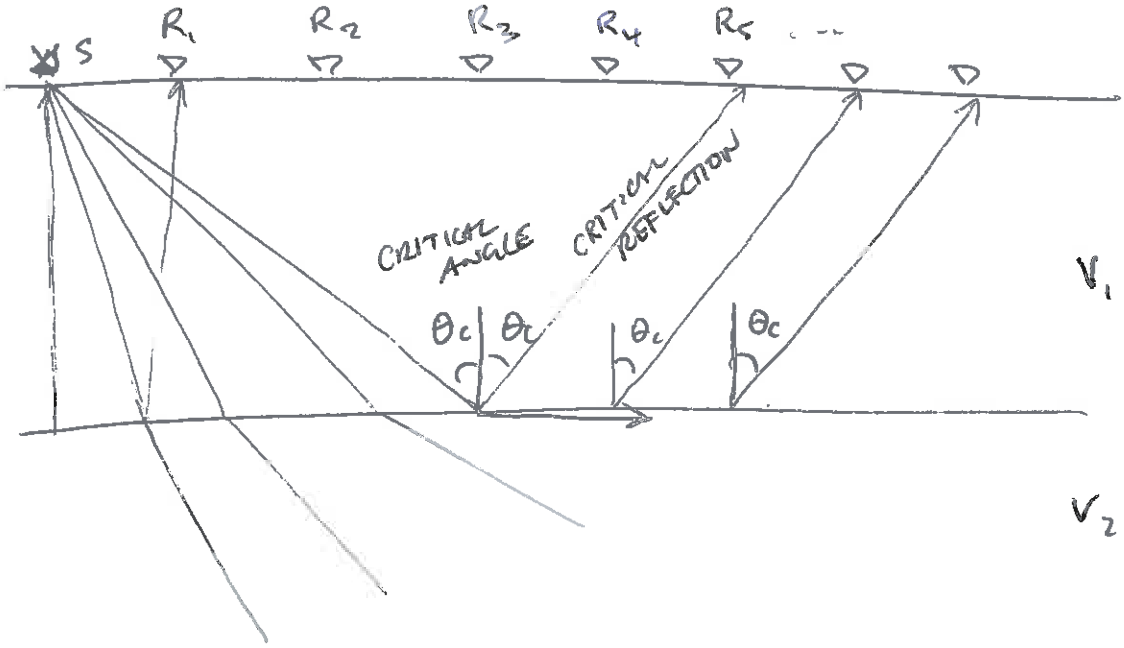


t_c
S
DISTANCE
DIRECT
REFRACTION
RELECTION
GROUND ROLL
AIR
time

TRAVEL-TIME DIAGRAM







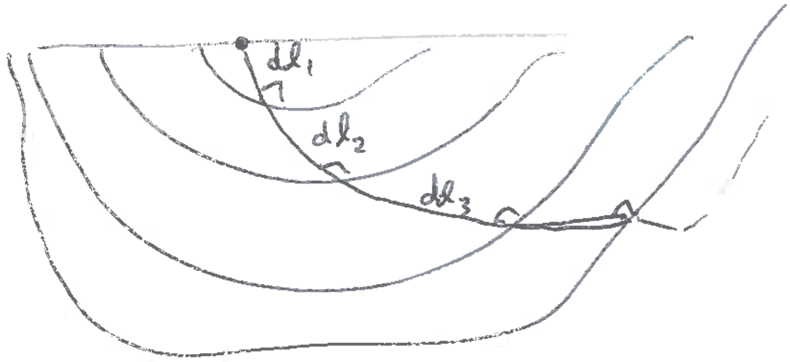
$$t = \frac{l}{v}$$

LENGTH OF PATH.

VELOCITY

travel time

$$t = \min \int_S^R \frac{dl}{v(x,y,z)}$$

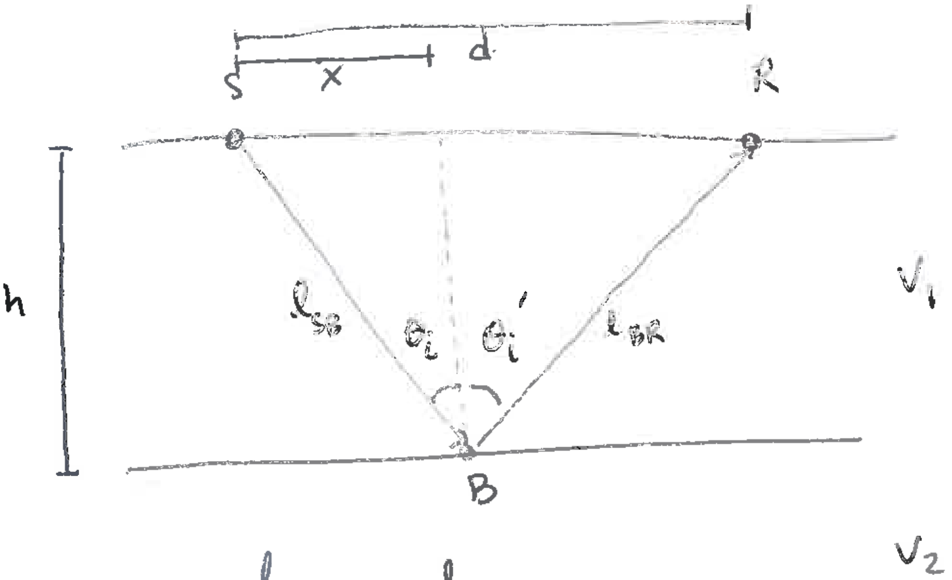


REFLECTION

ISOTROPIC

NO VELOCITY GRADIENTS

HORIZONTAL INTERFACES



$$t = \frac{l_{SB}}{V_1} + \frac{l_{BR}}{V_1}$$

$$l_{SB} = (x^2 + h^2)^{1/2} \qquad l_{BR} = ((d-x)^2 + h^2)^{1/2}$$

$$\min\left[t = \frac{(x^2+h^2)^{1/2}}{V_1} + \frac{((d-x)^2+h^2)^{1/2}}{V_1}\right]$$

$$\frac{dt}{dx} = 0 = \frac{1}{V_1}\left[\frac{x}{(x^2+h^2)^{1/2}} - \frac{(d-x)}{((d-x)^2+h^2)^{1/2}}\right]$$

$$\frac{d-x}{((d-x)^2+h^2)^{1/2}} = \frac{x}{(x^2+h^2)^{1/2}}$$

$$\frac{(d-x)^2}{(d-x)^2+h^2} = \frac{x^2}{x^2+h^2}$$

$$(x^2+h^2)(d-x)^2 = x^2((d-x)^2+h^2)$$

$$(x^2+h^2)(d^2-2xd+x^2) = x^2(d^2-2xd+x^2+h^2)$$

$$d^2h^2-2xdh^2+\cancel{x^2h^2} = \cancel{x^2h^2}$$

$$d^2h^2-2xdh^2 = 0$$

$$d\cancel{dh^2} = 2x\cancel{dh^2}$$

$$d = 2x$$

$$\Rightarrow \theta_i = \theta_i'$$

$$t = \frac{(x^2+h^2)^{1/2}}{v_1} + \frac{(x^2+h^2)^{1/2}}{v_1}$$

$$= \frac{2(x^2+h^2)^{1/2}}{v_1}$$

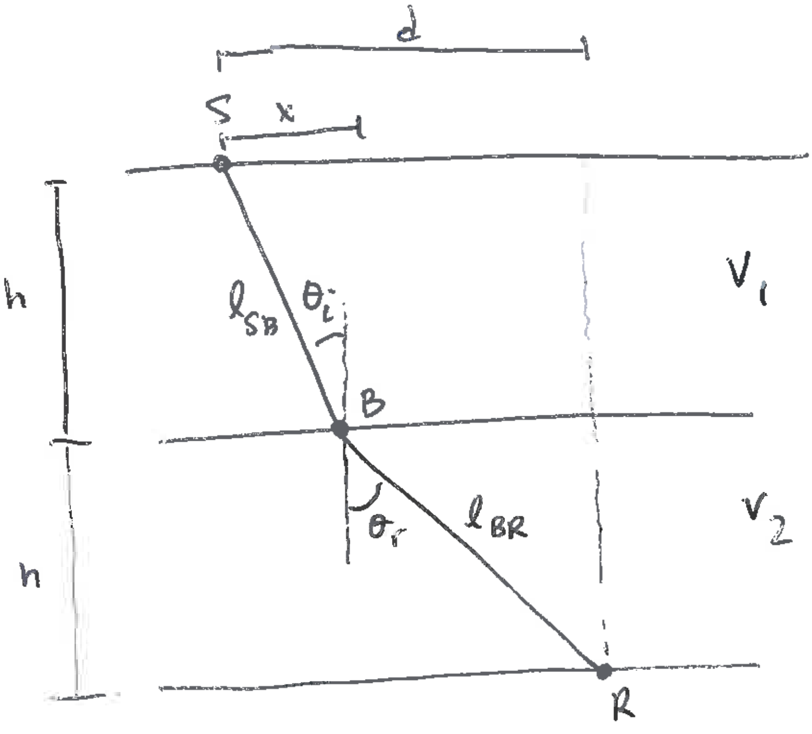
$$t = \frac{2h}{v_1}\left(1+\frac{x^2}{(2h)^2}\right)^{1/2} \Rightarrow t = t_0\left(1+\frac{x^2}{(t_0v_1)^2}\right)^{1/2}$$

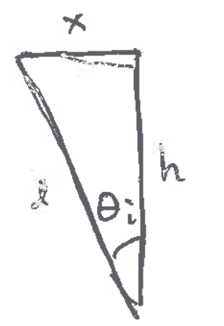
$$\frac{2h}{v_1}$$

$$1 = \underbrace{\frac{t^2}{(2hv_1^{-1})^2}}_{=a^2} - \underbrace{\frac{x^2}{(2h)^2}}_{=b^2} \quad \text{HYPERBOLA}$$

SLOPE OF ASYMPTOTES $\pm\frac{a}{b} = \frac{\cancel{2h}v_1^{-1}}{\cancel{2h}} = \frac{1}{v_1}$

$$t_i(x=0) = \frac{2h}{v_1}$$





$$l^2 = h^2 + x^2$$

$$\sin\theta_i = \frac{x}{l} = \frac{x}{(h^2+x^2)^{1/2}}$$

$$t = \frac{l_{SB}}{V_1} + \frac{l_{BR}}{V_2}$$

$$l_{SB} = (x^2+h^2)^{1/2} \qquad l_{BR} = ((d-x)^2 + h^2)^{1/2}$$

$$\min\left[t = \frac{(x^2+h^2)^{1/2}}{V_1} + \frac{((d-x)^2+h^2)^{1/2}}{V_2} \right]$$

$$\frac{dt}{dx} = 0 = \frac{1}{V_1}\left(\frac{x}{(h^2+x^2)^{1/2}}\right) - \frac{1}{V_2}\left(\frac{d-x}{(h^2+(d-x)^2)^{1/2}}\right)$$

$$= \frac{1}{V_1}\sin\theta_i - \frac{1}{V_2}\sin\theta_r$$

$$\frac{\sin\theta_r}{V_2} = \frac{\sin\theta_i}{V_1} \Rightarrow \text{SNELL'S LAW}$$

CRITICAL ANGLE, θ_c & $\theta_r = 90°$

$$\frac{1}{V_2} = \frac{\sin\theta_c}{V_1}$$

$$\sin\theta_c = \frac{V_1}{V_2}$$

RELECTION / TRANSMISSION COEFFICIENTS

NORMAL INCIDENT WAVE

$$R = \frac{Z_2 - Z_1}{Z_2 + Z_1}$$

Z_1 ← ACOUSTIC IMPEDANCE

$Z = \rho V$

ρ: DENSITY, V: VELOCITY.

R: REFLECTION COEFFICIENT

$$R = \frac{\rho_2 V_2 - \rho_1 V_1}{\rho_2 V_2 + \rho_1 V_1}$$

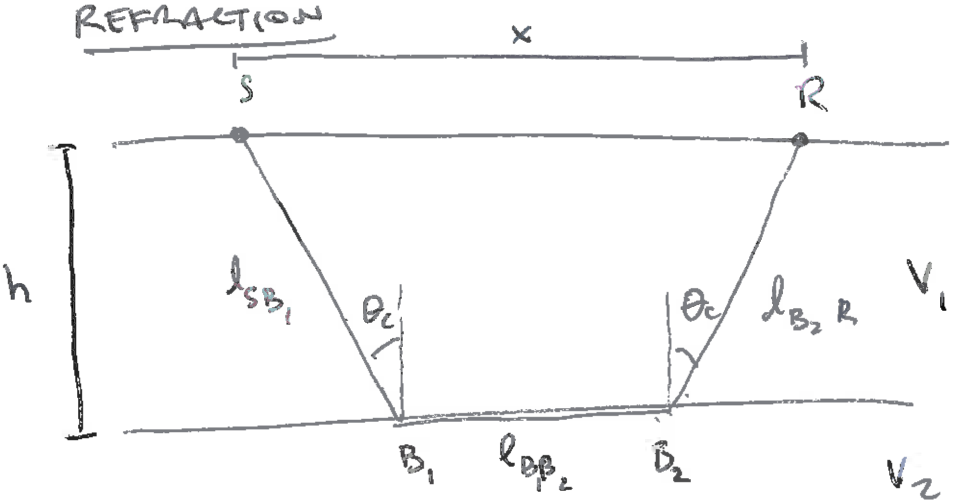
$R \to 1$ ALL ENERGY REFLECTED

$R \to 0$ ALL ENERGY IS TRANSMITTED

$$T = \frac{2Z_1}{Z_2 + Z_1}$$

T: TRANSMISSION COEFFICIENT

REFRACTION



$$t = \frac{l_{SB_1}}{V_1} + \frac{l_{B_1B_2}}{V_2} + \frac{l_{B_2R}}{V_1}$$